

Shaan Patel

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OBJECTIVE

Ambitious third-year Computer Engineering student specializing in RTL design, hardware/PCB prototyping, and embedded C programming. Seeking a firmware design or hardware design internship to apply expertise in SystemVerilog RTL implementation, embedded interfaces (SPI/I2C), and automated hardware testing to create high performance silicon systems.

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Engineering

Master of Science in Electrical and Computer Engineering

Atlanta, GA

August 2023 – May 2027

August 2027 – May 2028

- Minor: Applications of Artificial Intelligence and Machine Learning | GPA: 3.57
- Relevant Coursework: Data Structures and Algorithms, Embedded Systems Design, Advanced Microarchitecture Design
- Leadership: Captain of GT Ramblin' Raas, GTXR Executive Board, SiliconJackets Digital Design Team Lead

TECHNICAL SKILLS

Programming: SystemVerilog, Verilog, VHDL, RISC-V, C, C++, OpenGL, Python, Java, JavaScript, Tcl, Perl, Shell Scripting

Hardware: FPGA, ESP32, IC Breadboards, Oscilloscope, Raspberry Pi, Logic Analyzer, NVM, GPIO interrupts, I2C, SPI, UART

Frameworks: FreeRTOS, ARM MBED, Arduino, Pandas, SFML, CUDA, PyTorch, AWS (S3, DynamoDB, EC2), Kubernetes

Verification: Synopsys VCS, UVM, SVA, Coverage Analysis, RTL Simulation & Debug, Logic Synthesis, P&R, Multimeters

Platforms: Linux (Ubuntu, Slackware, Debian), Red Hat (OpenStack, CloudForms), OpenVMS, Solaris, Windows, MacOS

Software: KiCAD, Altium, Cadence Virtuoso, Intel Altera Quartus II, LTspice, CMake, Verilator, gRPC, Blender, Autodesk Fusion 360/AutoCAD, SolidWorks, Onshape, Visual Studio Code, GitHub, Wireshark, NI LabVIEW, TestStand, Inkscape, Slack

EXPERIENCE

AeroCore-V – Hardware-Accelerated SoC for Autonomous UAVs

Atlanta, GA

SiliconJackets Project Lead

August 2025 – Present

- Designed RTL for an open-source RISC-V SoC in SystemVerilog with custom ALU extensions for single-cycle PID control math
- Integrated a direct-mapped, 4KB L1 cache handling high-frequency I2C/SPI GPS and IMU data streams to reduce stalls by 52% (measured via a custom Verilator cycle-accurate harness), while developing a 4-way L2 controller with LRU replacement
- Developing a virtual memory system with Page Table Walker and TLB to isolate flight routines and protect from page faults, utilizing Python and Github Actions CI/CD pipelines to automate regression testing and cycle-accurate simulations
- Prototyped a priority-based thread scheduler to manage UAV flight loops and telemetry tasks, alongside an OpenGL Digital Twin physics simulator in C++ to validate future flight latency reductions
- Validated logic synthesis and PPA analysis on Intel DE10 FPGAs with Intel Quartus Prime and authored SystemVerilog Assertions (SVA) and functional testbenches in Verilator to perform coverage analysis and debug RTL logic failures

Georgia Tech HIVE Makerspace

Atlanta, GA

Student Leader and Volunteer

August 2024 – Present

- Instruct ~40 students per week on advanced design and fabrication workflows, including 3D CAD/slicing, circuit analysis with oscilloscopes, laser cutter operation, KiCAD/Altium PCB design and manufacturing (taking a blank FR4 to an operational PCB)
- Deployed an open-source multi-device print manager to automate jobs on a FIFO basis, increasing the operational uptime of the free-to-use 12-unit Bambu Lab 3D print farm for the Georgia Tech student body
- Facilitate workshops and created training resources that contributed to a 20% increase in student participation
- Architecting a full-stack inventory management system using an AWS RDS backend to track materials in the cloud with S3 and DynamoDB to enable a student electronics rental program, aiming to reduce annual student costs

ESP32 Fiber Optic Star Map

Atlanta, GA

Sole Creator

September 2025 – December 2025

- Scripted bare-metal C++ firmware for an ESP32 microcontroller, implementing non-volatile memory (NVM) drivers for state retention and hardware-debounced GPIO interrupt service routines (ISRs) for real-time control via WiFi wireless communication
- Engineered ARGB analog PWM power linkage networks and modeled a custom enclosure in SolidWorks for thermal dissipation
- Fabricated a custom ESP32 PCB designed with KiCad using a ProtoLaser, running SPICE circuit simulations and passing DRC and ERC verification to ensure clean trace routing and high frequency signal integrity for seamless hardware software integration
- Executed hardware validation using oscilloscopes and multimeters to characterize transient voltage drop under 100% LED load

Academy of Science International Research

Ashburn, VA

Research Collaborator

November 2022 – July 2024

- Synthesized a novel saline-soluble polymer alternative to general petrochemical plastics as a means of preventing marine pollution
- Modeled and fabricated 3D-generated molds using filament and resin printers for synthesizing hydrogels through incubation
- Worked with a LabVIEW tensile tester to measure Young's modulus and SEM electron microscopy to analyze sample composition
- Led collaboration efforts with South Korean research teams for two years to formulate samples and complete testing
- Collected over 150 trials of data and was awarded the Yale Outstanding Project award at RSEF in Spring of 2023